# GM-6 - Singleton AudioSettings

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UML contrastado con el arbol sintactico y la resolucion semantica de Java. El documento separa la vista conectada de relaciones y los compartimentos completos de cada clase para mantener legibilidad.

**Problema.** Evitar preferencias contradictorias de volumen y mute entre el tema del menu y la banda sonora de la partida.

**Contexto.** MainMenuScreen y SoundtrackPlayer mantienen reproductores distintos; ambos deben respetar una configuracion unica durante el proceso.

**Solucion real.** AudioSettings es final, tiene constructor privado y una sola INSTANCE static final, accesible con getInstance. Menu y reproductor conservan esa referencia. GameScreen consulta las preferencias existentes o asigna parametros explicitamente antes de crear el reproductor. Ambos audios sincronizan el volumen efectivo; sus caches no constituyen preferencias independientes.

**Persistencia.** Recrear pantallas y reiniciar la partida conserva las preferencias. No se afirma persistencia entre procesos ni se introduce GameSession.

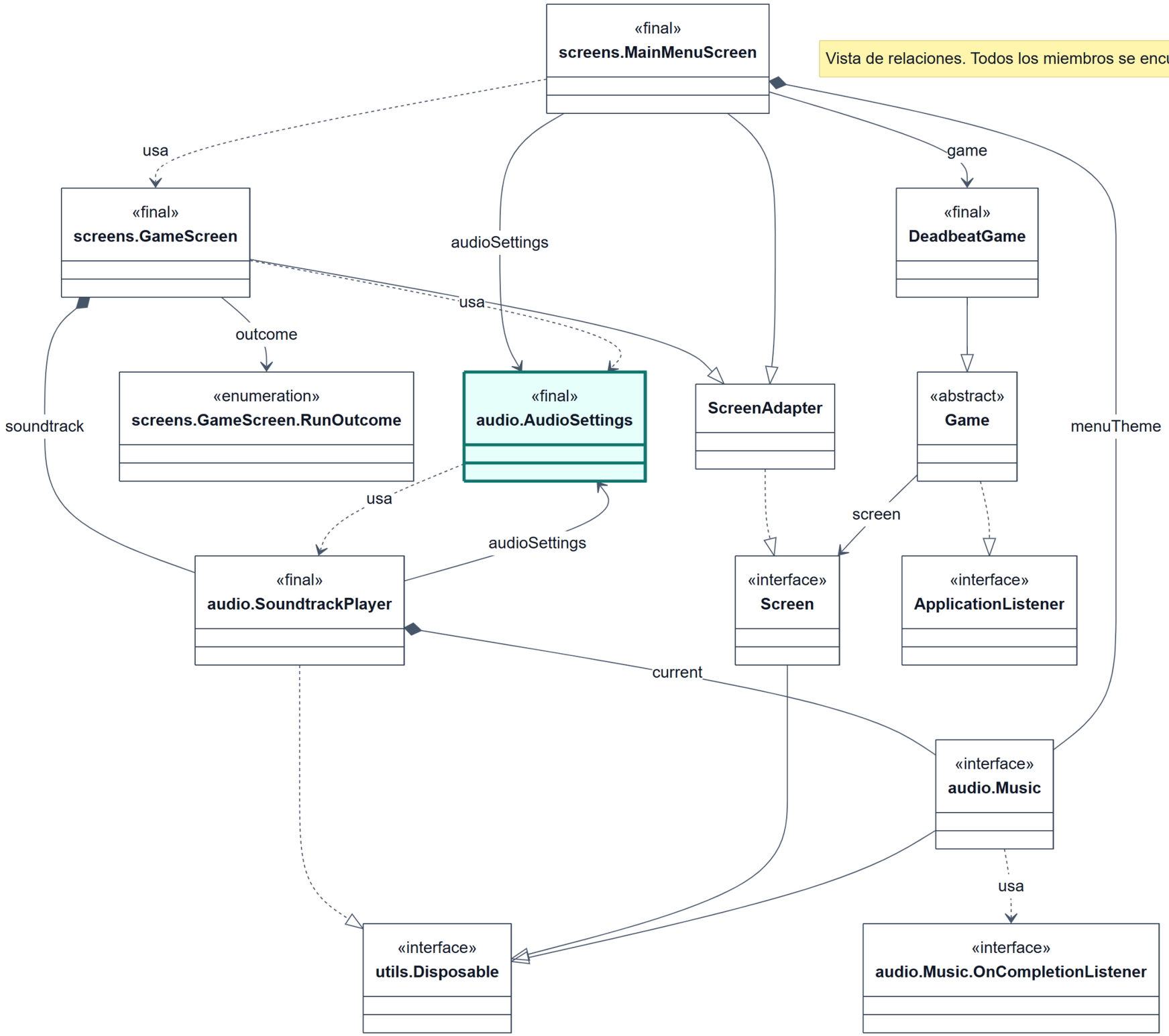
**Lectura.** + publico; - privado; # protegido; ~ acceso de paquete. Miembros static subrayados, metodos abstractos en cursiva. Las restricciones se escriben entre llaves; no se copian miembros heredados ni se inventan constructores implicitos.

**Relaciones.** Triangulo continuo: herencia; triangulo discontinuo: realizacion; flecha continua: asociacion; flecha discontinua: dependencia; rombo lleno: composicion. Las composiciones se basan en creacion y propiedad del objeto, no solo en final.

**Alcance externo.** Se incluyen completas las pequenas bases de libGDX que enlazan las pantallas, el juego y Music. FileHandle, las clases del launcher, LinkedHashMap y AutoCloseable son referencias externas de frontera. No se expande recursivamente toda la biblioteca.

**Advertencia de formato.** El maestro PNG es un poster de consulta con zoom. Para Google Docs se incluyen imagenes individuales de cada clase y secciones. Las paginas de miembros de este PDF son A4 y mantienen texto legible, continuando las clases extensas en la pagina siguiente.

# Vista conectada de relaciones



# Trazabilidad Java / UML

A: atributos. C: constructores explicitos. M: metodos. Cada constante enum cuenta como atributo. Cada sobrecarga cuenta por separado. Los valores J/U comparan Java con el UML maestro.

| Clase / interfaz | A J/U | C J/U | M J/U | Dif. |
|---|---|---|---|---|
| audio.AudioSettings | 3 / 3 | 1 / 1 | 7 / 7 | 0 |
| audio.SoundtrackPlayer | 16 / 16 | 2 / 2 | 22 / 22 | 0 |
| DeadbeatGame | 0 / 0 | 0 / 0 | 2 / 2 | 0 |
| screens.GameScreen | 70 / 70 | 2 / 2 | 36 / 36 | 0 |
| screens.GameScreen.RunOutcome | 3 / 3 | 0 / 0 | 0 / 0 | 0 |
| screens.MainMenuScreen | 30 / 30 | 1 / 1 | 17 / 17 | 0 |
| ScreenAdapter | 0 / 0 | 0 / 0 | 7 / 7 | 0 |
| Screen | 0 / 0 | 0 / 0 | 7 / 7 | 0 |
| Game | 1 / 1 | 0 / 0 | 7 / 7 | 0 |
| ApplicationListener | 0 / 0 | 0 / 0 | 6 / 6 | 0 |
| utils.Disposable | 0 / 0 | 0 / 0 | 1 / 1 | 0 |
| audio.Music | 0 / 0 | 0 / 0 | 13 / 13 | 0 |
| audio.Music.OnCompletionListener | 0 / 0 | 0 / 0 | 1 / 1 | 0 |

# Clases propias de produccion

## audio.AudioSettings <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/AudioSettings.java:4
public final

ATRIBUTOS

-INSTANCE: AudioSettings {static, final}
-volume: float
-muted: boolean

CONSTRUCTORES EXPLICITOS

-AudioSettings()

METODOS DECLARADOS

+getInstance(): AudioSettings {static}
+getVolume(): float
+setVolume(float volume): void
+isMuted(): boolean
+setMuted(boolean muted): void
+toggleMute(): void
+getEffectiveVolume(): float

Singleton: constructor privado, INSTANCE static final y getInstance static. Las preferencias pertenecen al objeto.

## audio.SoundtrackPlayer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/SoundtrackPlayer.java:27
public final

ATRIBUTOS

-MAX_TRACK_NUMBER: int {static, final}
-MAX_HISTORY: int {static, final}
+DEFAULT_VOLUME: float {static, final}
-random: Random {final}
-trackMetadata: Properties {final}
-history: List<Integer> {final}
-historyPosition: int
-order: int[]
-nextPosition: int
-lastPlayedIndex: int
-current: Music
-currentIndex: int
-appliedVolume: float
-audioSettings: AudioSettings {final}
-nextRequested: boolean {volatile}
-disposed: boolean

CONSTRUCTORES EXPLICITOS

+SoundtrackPlayer()
~SoundtrackPlayer(Random random)

METODOS DECLARADOS

+update(): void
+nextTrack(): void
+previousTrack(): void
-playFromHistory(int position, int direction): boolean
-playNextFresh(): void
-playIndex(int index): boolean
-pathOf(int index): String {static}

## audio.SoundtrackPlayer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/SoundtrackPlayer.java:27
public final

-shuffleNewCycle(): void

-loadMetadata(): void

+getCurrentTitle(): String

+getCurrentArtist(): String

-metadataKey(int index, String field): String {static}

+getTrackNumber(): int

+getAvailableTrackCount(): int

+getVolume(): float

+isMuted(): boolean

+setVolume(float amount): void

+toggleMute(): void

-effectiveVolume(): float

-applyVolume(): void

-stopCurrent(): void

+dispose(): void

Implements: Disposable. appliedVolume es solo la cache del valor enviado a Music; AudioSettings conserva las preferencias.

## DeadbeatGame <<class>>

DeadBeat/core/src/main/java/com/deadbeat/DeadbeatGame.java:8
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+create(): void

+dispose(): void

Extends: Game.

## screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

ATRIBUTOS

+VIEW_WIDTH: float {static, final}

+VIEW_HEIGHT: float {static, final}

-GUITAR_X: float {static, final}

-GUITAR_WIDTH: float {static, final}

-GUITAR_HEIGHT: float {static, final}

-GUITAR_Y: float {static, final}

-MASK_WIDTH: int {static, final}

-MASK_HEIGHT: int {static, final}

-PLAYER_HUD_W: float {static, final}

-PLAYER_HUD_H: float {static, final}

-PLAYER_HUD_X: float {static, final}

-PLAYER_HUD_Y: float {static, final}

-MUSIC_BUTTON_W: float {static, final}

-MUSIC_BUTTON_H: float {static, final}

-MUSIC_BUTTON_Y: float {static, final}

-PREVIOUS_X: float {static, final}

## screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

-NEXT_X: float {static, final}
-MUSIC_HUD_HOLD_SECONDS: float {static, final}
-MUSIC_HUD_FADE_SECONDS: float {static, final}
-RESULT_X: float {static, final}
-RESULT_Y: float {static, final}
-RESULT_W: float {static, final}
-RESULT_H: float {static, final}
-RESULT_BUTTON_X: float {static, final}
-RESULT_BUTTON_W: float {static, final}
-RESULT_BUTTON_H: float {static, final}
-RESULT_RETRY_Y: float {static, final}
-RESULT_EXIT_Y: float {static, final}
-MENU_X: float {static, final}
-MENU_Y: float {static, final}
-MENU_WIDTH: float {static, final}
-MENU_HEIGHT: float {static, final}
-BUTTON_X: float {static, final}
-BUTTON_WIDTH: float {static, final}
-BUTTON_HEIGHT: float {static, final}
-CONTINUE_Y: float {static, final}
-EXIT_Y: float {static, final}
-SPEAKER_X: float {static, final}
-SPEAKER_Y: float {static, final}
-SPEAKER_SIZE: float {static, final}
-VOLUME_X: float {static, final}
-VOLUME_Y: float {static, final}
-VOLUME_WIDTH: float {static, final}
-VOLUME_HEIGHT: float {static, final}
-worldViewport: FitViewport {final}
-hudViewport: FitViewport {final}
-batch: SpriteBatch {final}
-titleFont: BitmapFont {final}
-menuFont: BitmapFont {final}
-smallFont: BitmapFont {final}
-trackFont: BitmapFont {final}
-glyph: GlyphLayout {final}
-mouseInHud: Vector2 {final}
-pixel: Texture {final}
-guitarMask: Texture {final}
-guitarOutline: Texture {final}
-world: GameWorld {final}
-soundtrack: SoundtrackPlayer {final}
-outcome: RunOutcome
-selectedResultButton: int
-runEndedAtNanos: long
-finalWave: int
-paused: boolean
-draggingVolume: boolean

## screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

-selectedButton: int

-displayedTrackNumber: int

-displayedTitle: String

-displayedArtist: String

-musicHudChangedAtNanos: long

-runStartedAtNanos: long

CONSTRUCTORES EXPLICITOS

+GameScreen()

+GameScreen(float initialVolume, boolean initiallyMuted)

METODOS DECLARADOS

+render(float delta): void

-drawExperienceBar(): void

-handleUpgradeInput(): void

-drawUpgradeSelection(): void

-checkRunOutcome(): void

-finishRun(RunOutcome result): void

-resetRun(): void

-handleResultInput(): void

-resultButtonUnderMouse(float x, float y): int

-activateResultButton(int button): void

-handleMusicHudClick(): void

-handlePauseMenuInput(): void

-changeVolumeAt(float mouseX): void

-buttonUnderMouse(float x, float y): int

-inside(float x, float y, float left, float bottom, float w, float h): boolean {static}

-activateButton(int button): void

-followPlayer(): void

-drawGuitarHealth(Player player): void

-syncMusicHudTrack(): void

-musicHudOpacity(): float

-drawMusicHud(): void

-musicBlock(float x, float y, float w, float h, float r, float g, float b, float baseAlpha, float fade): void

-drawMusicHudButton(float x, float y, String label, float fade): void

-fitting(BitmapFont font, String value, float maxWidth): String

-drawResultScreen(): void

-drawResultButton(float y, String label, boolean active, float ar, float ag, float ab): void

-drawPauseMenu(): void

-drawVolumeControl(): void

-drawPauseButton(float y, String label, boolean selected): void

-elapsedAsClock(): String

-two(long value): String {static}

-centered(BitmapFont font, String text, float centerX, float baseline, float r, float g, float b, float a): void

-block(float x, float y, float w, float h, float r, float g, float b, float a): void

-clamp(float v, float lo, float hi): float {static}

+resize(int width, int height): void

+dispose(): void

## screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

Extends: ScreenAdapter. Los dos constructores respetan la configuracion compartida o asignan argumentos explicitamente antes de reproducir.

## screens.GameScreen.RunOutcome <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:60
private static final

ATRIBUTOS

+PLAYING: RunOutcome {static, final}

+DEFEAT: RunOutcome {static, final}

+VICTORY: RunOutcome {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.screens.GameScreen. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

## screens.MainMenuScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/MainMenuScreen.java:24
public final

ATRIBUTOS

-W: float {static, final}

-H: float {static, final}

-BUTTON_X: float {static, final}

-BUTTON_W: float {static, final}

-BUTTON_H: float {static, final}

-PLAY_Y: float {static, final}

-EXIT_Y: float {static, final}

-THEME_PATH: String {static, final}

-VOLUME_X: float {static, final}

-VOLUME_Y: float {static, final}

-VOLUME_W: float {static, final}

-VOLUME_H: float {static, final}

-SPEAKER_X: float {static, final}

-SPEAKER_Y: float {static, final}

-SPEAKER_SIZE: float {static, final}

-game: DeadbeatGame {final}

-viewport: FitViewport {final}

-batch: SpriteBatch {final}

-titleFont: BitmapFont {final}

-menuFont: BitmapFont {final}

-glyph: GlyphLayout {final}

-mouse: Vector2 {final}

-pixel: Texture {final}

-menuTheme: Music

-selected: int

-elapsed: float

-disposed: boolean

-draggingVolume: boolean

-audioSettings: AudioSettings {final}

### screens.MainMenuScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/MainMenuScreen.java:24
public final

-appliedThemeVolume: float

CONSTRUCTORES EXPLICITOS

+MainMenuScreen(DeadbeatGame game)

METODOS DECLARADOS

+render(float delta): void

-handleInput(): boolean

-buttonAt(float x, float y): int

-inside(float x, float y, float bx, float by, float bw, float bh): boolean {static}

-activate(int choice): boolean

-drawScenery(): void

-drawMenu(): void

-drawVolumeSlider(): void

-changeVolume(float value): void

-toggleMute(): void

-applyThemeVolume(): void

-drawButton(float y, String label, boolean active): void

-centered(BitmapFont font, String text, float cx, float baseline, float r, float g, float b, float alpha): void

-box(float x, float y, float w, float h, float r, float g, float b, float alpha): void

-stopTheme(): void

+resize(int width, int height): void

+dispose(): void

Extends: ScreenAdapter. No declara volume ni muted independientes; usa AudioSettings y una cache de aplicacion.

# Bases e interfaces de apoyo de libGDX

### ScreenAdapter <<class>>

libGDX 1.14.2 sources / ScreenAdapter.java
public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+render(float delta): void

+resize(int width, int height): void

+show(): void

+hide(): void

+pause(): void

+resume(): void

+dispose(): void

## Screen <<interface>>

libGDX 1.14.2 sources / Screen.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

*+show(): void {abstract}*

*+render(float delta): void {abstract}*

*+resize(int width, int height): void {abstract}*

*+pause(): void {abstract}*

*+resume(): void {abstract}*

*+hide(): void {abstract}*

*+dispose(): void {abstract}*

## Game <<class>>

libGDX 1.14.2 sources / Game.java
public abstract

ATRIBUTOS

#screen: Screen

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+dispose(): void

+pause(): void

+resume(): void

+render(): void

+resize(int width, int height): void

+setScreen(Screen screen): void

+getScreen(): Screen

## ApplicationListener <<interface>>

libGDX 1.14.2 sources / ApplicationListener.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

*+create(): void {abstract}*

*+resize(int width, int height): void {abstract}*

*+render(): void {abstract}*

*+pause(): void {abstract}*

*+resume(): void {abstract}*

*+dispose(): void {abstract}*

## utils.Disposable <<interface>>

libGDX 1.14.2 sources / Disposable.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

*+dispose(): void {abstract}*

## audio.Music <<interface>>

libGDX 1.14.2 sources / Music.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

*+play(): void {abstract}*

*+pause(): void {abstract}*

*+stop(): void {abstract}*

*+isPlaying(): boolean {abstract}*

*+setLooping(boolean isLooping): void {abstract}*

*+isLooping(): boolean {abstract}*

*+setVolume(float volume): void {abstract}*

*+getVolume(): float {abstract}*

*+setPan(float pan, float volume): void {abstract}*

*+setPosition(float position): void {abstract}*

*+getPosition(): float {abstract}*

*+dispose(): void {abstract}*

*+setOnCompletionListener(OnCompletionListener listener): void {abstract}*

## audio.Music.OnCompletionListener <<interface>>

libGDX 1.14.2 sources / OnCompletionListener.java
abstract public static

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

*+onCompletion(Music music): void {abstract}*

Tipo anidado en com.badlogic.gdx.audio.Music.